

Homework (Habanero)

- Q1.** A patient needs to take $3x+2$ milligrams of medicine, where x is the number of doses. If the patient can take at most 15 milligrams per day, what is the solution to the inequality $3x+2 \leq 15$?
- (A) $x \leq 3$
(B) $x \leq 4$
(C) $x \leq 5$
(D) $x < 4$
(E) $x > 4$
- Q2.** A biologist is studying the growth of a cell population, which is modeled by the inequality $2x-5 > 3$. What is the solution to this inequality?
- (A) $x > 2$
(B) $x > 3$
(C) $x > 4$
(D) $x < 2$
(E) $x < 3$
- Q3.** The concentration of a medicine in the bloodstream is given by the inequality $0.5x+2 \geq 4$, where x is the time in hours. What is the solution to this inequality?
- (A) $x \geq 2$
(B) $x \geq 3$
(C) $x \geq 4$
(D) $x \leq 2$
(E) $x \leq 3$
- Q4.** A medical trial requires that the participants' blood pressure be within the range $80 \leq 2x-3 \leq 120$. What is the solution to this compound inequality?
- (A) $\frac{83}{2} \leq x \leq \frac{123}{2}$
(B) $\frac{81}{2} \leq x \leq \frac{123}{2}$
(C) $\frac{83}{2} \leq x \leq \frac{125}{2}$
(D) $\frac{81}{2} \leq x \leq \frac{125}{2}$
(E) $\frac{83}{2} \leq x \leq \frac{121}{2}$
- Q5.** The graph of the inequality $x+2y > 4$ is a region in the xy -plane. Which of the following points lies within this region?
- (A) (0,0)
(B) (0,2)
(C) (2,0)
(D) (1,1)
(E) (3,3)
- Q6.** The number of bacteria in a culture is modeled by the inequality $3x-2 > 5$, where x is the time in days. What is the solution to this inequality in interval notation?
- (A) $(2, \infty)$
(B) $(3, \infty)$
(C) $(\frac{7}{3}, \infty)$
(D) $(-\infty, 2)$
(E) $(-\infty, 3)$
- Q7.** A doctor prescribes a patient to take x milligrams of medicine, where $2x+5 \leq 20$. What is the maximum amount of medicine the patient can take?
- (A) 5 milligrams
(B) 7 milligrams
(C) 7.5 milligrams
(D) 10 milligrams
(E) 12 milligrams
- Q8.** A patient's heart rate is monitored, and it must satisfy the inequality $4x-3 \geq 2$, where x is the time in hours. What is the solution to this inequality?
- (A) $x \geq 1$
(B) $x \geq 1.25$
(C) $x \geq 2$
(D) $x \leq 1$
(E) $x \leq 1.25$

Open Ended Questions (FRQ)

- Q9.** The concentration of a certain medicine in the bloodstream is modeled by the inequality $0.2x+1.5 \leq 3.2$, where x is the time in hours. Solve for x and graph the solution on a number line. Write the solution in interval notation.
- Q10.** A biologist is studying the growth of a cell population. The population grows at a rate that satisfies the inequality $2x-5 \geq x+2$, where x is the time in days. Solve for x and write the solution in interval notation. Provide a brief explanation of the biological significance of this inequality.

Chef's Tips

- Provide clear instructions and examples for each question type
- Ensure that students have access to necessary resources and materials
- Encourage students to use graph paper for graphing inequalities